Relational Data Concepts

Estimated time needed: 10 minutes

In this lab, you will apply the concepts that you learnt in this module to a relational database schema called Car Dealership, which is designed to keep track of automobile sales in a car dealership.

Objectives:

After completing this lab, you will be able to:

- Evaluate your knowledge of Relational Database Concepts and the Entity-Relationship (ER) Diagram
- Imporve your understanding of terms related to relational models like entity, attribute, and keys.

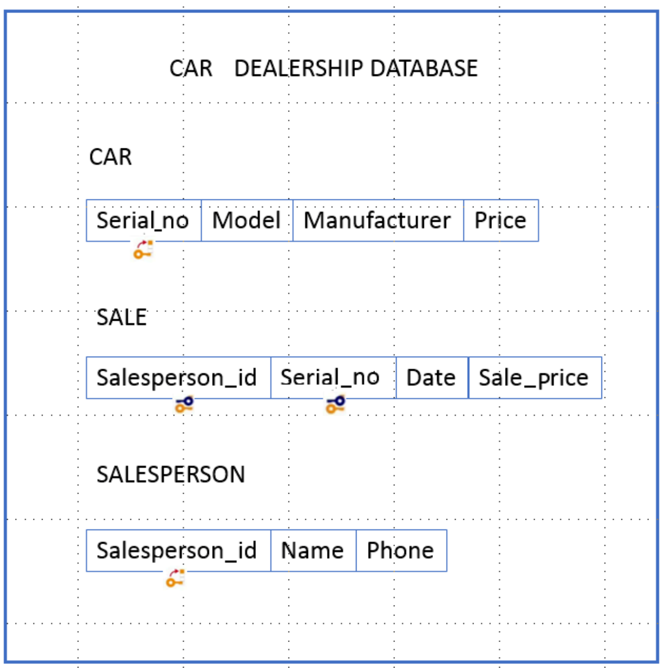
Concepts covered in the lab

1. `Entity`: A person, place, object, or event about which data is stored (represented as a table in a database).
2. `Attribute`: A property or characteristic of an entity (represented as a column in a table).
3. `Primary Key`: A unique identifier for each row in a table.
4. `Foreign Key`: A field in one table that refers to the primary key of another table, creating a relationship between them.
5. `Entity-Relationship (ER) Diagram`: A diagram that shows entities (tables), attributes (columns), and the relationships between them.

Exercise

In this exercise, we will be working on a relational database schema called Car Dealership. A database has to be designed to keep track of automobile sales in a car dealership.

Schema diagram for the Car Dealership relational database:



Relational instance of SALE:

Salesperson_id	Serial_no	Date	Sale_price
10001	1we4ds87	12/03/2020	$ 10,000.00
10005	d63jw3ty	12/03/2020	$ 5,000.00
10009	sy63bjd1	13/03/2020	$ 25,000.00
10001	k2k4edr8	13/03/2020	$ 49,000.00
10051	w3r334ac	13/03/2020	$ 8,000.00

Now let us go through some questions based on the above database schema of Car Dealership and relational instance of SALE:

1. How many relations does the Car Dealership database schema contain?

   ▼ Hint

   A relation is also the mathematical term for a table.

   ▼ Answer

   3. The Car Dealership database schema contains the following 3 relations or tables: CAR, SALE, SALESPERSON.

2. How many columns does the relation Car contain?

   ▼ Hint

   A relation is also the mathematical term for a table. A table is a combination of rows and columns. The columns are the attributes, or fields.

   ▼ Answer

   4. The relation Car contains the following 4 columns: Serial_no, Model, Manufacturer, Price.

3. How many rows does the relation Sale contain?

   ▼ Hint

   A relation is also the mathematical term for a table. A table is a combination of rows and columns. The rows are the tuples.

   ▼ Answer

   5

4. Identify the attributes of the relation Salesperson.

   ▼ Hint

   A Relational Schema specifies the relation name and type of each of the columns, which are the attributes.

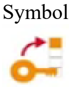
   ▼ Answer

   Salesperson_id, Name, Phone

5. Identify which relations of the Car Dealership database have primary keys. Name the primary keys if exist.

   ▼ Hint

   A primary key is a column or group of columns that uniquely identify every row in a relation/table. A table cannot have more than one primary key. Symbol representing a primary key may look like:

   ▼ Answer

   The relations **CAR** and **SALESPERSON** have primary keys. **Serial_no** is the primary key of the relation CAR. **Salesperson_id** is the primary key of the relation SALESPERSON.

6. Which table(s) in the Car Dealership database include foreign keys? Identify the foreign keys and describe the table and column they reference.

   ▼ Hint

   A foreign key is a column that establishes a relationship between two relations/tables. It acts as a cross-reference between two tables as it points to the primary key of another table. A table can have multiple foreign keys referencing primary keys of other tables. Symbol representing a foreign key may look like:

   ▼ Answer

   The relation **SALE** has foreign keys. **Serial_no** is a foreign key of the relation SALE which acts as a cross-reference between two relations CAR and SALE as it points to the primary key of the relation CAR. **Salesperson_id** is another foreign key of the relation SALE which acts as a cross-reference between two relations CAR and SALESPERSON as it points to the primary key of the relation SALESPERSON.

Summary

Congratulations!

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